In [ ]:
```
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```

In [ ]:
```
#Week 1 Demo
```

In [3]:
```
print("Hello, World!")
```

```
Hello, World!
```

In [1]:
```
a=2
b=3
c=a+b
print("c=",c)
```

```
c= 5
```

In [2]:
```
print(a/b)
print(a//b)
```

```
0.6666666666666666
0
```

In [12]:
```
x=0.345
y=6.8e-4
print(x+y)
```

```
0.34568
```

In [4]:
```
str_a="This is my first Python class"
str_b="""String that spans
Multiple Lines
"""
print(str_b)
str_b.count("\n")
```

```
String that spans
Multiple Lines
```

Out[4]: 2

In [5]:
```
str_c=str_a.replace("class","demonstration")
str_c
```

Out[5]: 'This is my first Python demonstration'

In [6]:
```
bool_a=True
bool_b=False
bool_a?
```

In [7]:
```
int_=1
bool_c=bool(int_)
bool_c
print(type(int_))
print(type(bool_c))
```

```
<class 'int'>
<class 'bool'>
```

In [8]:
```
def fn(a,b,c=None):
    result=a+b
    if c is not None:
        resulresult*c
    return result
```

In [9]:
```
fn?
fn(1,2,None)
```

Out[9]: 3

In [11]:
```
x=-5
if x<0:
    print("It's negative")
```

```
It's negative
```

In [16]:
```
x=5
if x<0:
    print("It's negative")
elif x==0:
    print("Equal to zero")
elif 0<x<5:
    print("Positive but smaller than 5")
else:
    print("Positive and larger than or equal to 5")
```

```
Positive and larger than or equal to 5
```

In [18]:
```
a=[1,2,None,4,None,5]
tot=0
for i in a:
    if i is None:
        continue
    tot+=i
tot
```

Out[18]: 12

In [20]:
```
a=[1,2,0,4,6,5,2,1]
tot=0
for i in a:
    if i==5:
        break
    tot+=i
tot
```

Out[20]: 13

In [22]:
```
x=256
tot=0
while x>0:
    if tot>500:
        break
    tot+=x
    x=x//2
tot
```

Out[22]: 504

In [24]:
```
x=0
if x<0:
    print("negative!")
elif x==0:
    pass
else:
    print("positive!")
```

```
hi
```

In [25]:
```
list(range(10))
```

Out[25]: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9]

In [27]:
```
a=[1,2,3,4]
for i in range(len(a)):
    print(a[i])
```

```
1
2
3
4
```

In [ ]:
```
##Lab Manual
```

In [28]:

```python
#1. Write a Python function to input two numbers and perform the calcul
def calculations(x,y):
    print("Addition:",a+b)
    print("Subtraction:",a-b)
    print("Multiplication:",a*b)
    print("Diviion:",a/b)
a=int(input("Enter an integer:"))
b=int(input("Enter an integer:"))
calculations(a,b)
```

```
Enter an integer:7
Enter an integer:5
Addition: 12
Subtraction: 2
Multiplication: 35
Diviion: 1.4
```

In [7]:

```python
#2. Write a Python function that takes an integer and returns True if i
def prime(a):
    f=1
    for i in range(2,a):
        if a%i==0:
            f=0
            break
    if f==1:
        return True
    else:
        return False
x=int(input("Enter an integer:"))
print(prime(x))
```

```
Enter an integer:15
False
```

In [10]:

```
#3. Create a Python function that creates a sequence between 1 and 100 (
sum=0
for i in range(101):
    if i%2==0:
        sum+=i
    else:
        print(i)
print("Sum:",sum)
```

```
1
3
5
7
9
11
13
15
17
19
21
23
25
27
29
31
33
35
37
39
41
43
45
47
49
51
53
55
57
59
61
63
65
67
69
71
73
75
77
79
81
83
85
87
89
91
93
95
97
99
Sum: 2550
```

In [14]:

```python
#4. Write a Python function to add two elements and display the result.
def add(a,b):
    return a+b
x=input("Enter an element:")
y=input("Enter an element:")
print(add(x,y))
```

```
Enter an element:hello
Enter an element:world
hello world
```

In [2]:

```python
#5. Write a Python function that takes a string input from the user and
def countstring(a):
    v=0
    c=0
    for i in a:
        if i in ['a','e','i','o','u']:
            v=v+1
        else:
            c=c+1
    print("Vowel Count:",v)
    print("Consonant Count:",c)
x=input("Enter a string:")
countstring(x)
```

```
Enter a string:hello world
Vowel Count: 3
Consonant Count: 8
```